

Differential Privacy and Cooperation in Public Goods Games: A Game-Theoretic Analysis

This report studies how differential privacy affects strategic behavior in the Public Goods Game (PGG), a standard model of the free-rider problem. In repeated settings with complete information, cooperation is hard to sustain because players can observe deviations and respond by defecting, which leads to inefficient outcomes.

To address this, we introduce differential privacy into the feedback mechanism by adding Laplace noise to the total contributions observed by players. This turns the game into one with uncertainty, where agents have to form beliefs and make decisions based on noisy signals instead of exact information.

We find that the relationship between privacy and cooperation is non-monotonic. When noise is too high, the signal becomes unreliable and coordination breaks down. When noise is very low, players react strongly to small deviations, which can destabilize cooperation. But for moderate levels of noise, cooperation can be sustained because players don't overreact to minor fluctuations.

We also treat the privacy parameter ϵ as a design choice and study how it affects social welfare. From simulations, we see that moderate values of ϵ give near-optimal outcomes by balancing accuracy and stability.

Overall, this shows that differential privacy is not just useful for protecting data, it can also shape incentives and improve outcomes in strategic settings.

Introduction

Many multi-agent systems have a basic tension between individual incentives and what's best for the group. This shows up in things like resource sharing, collaborative data contribution, and distributed learning, where everyone benefits from a shared outcome but still has an incentive to free-ride.

The Public Goods Game (PGG) is a standard way to study this. In repeated versions of the game with complete information, cooperation is hard to maintain. Since players can directly observe when someone defects, even small deviations can trigger retaliation, which eventually leads to a breakdown of cooperation.

At the same time, many modern systems operate under privacy constraints. Methods like ϵ -differential privacy ensure that individual-level information can't be inferred from aggregate data. Because of this, agents don't see exact outcomes anymore, they see noisy signals.

This leads to an important question:

How does differential privacy affect behavior in settings with free-riding incentives?

In this project, we study a repeated Public Goods Game where total contributions are revealed through a differentially private mechanism. This introduces uncertainty, so agents have to form beliefs based on noisy observations instead of exact values.

The main idea is that privacy doesn't just reduce information, it also changes incentives. Too much noise makes coordination difficult, but a moderate amount of noise can actually help sustain cooperation by preventing players from overreacting to small changes.

We also look at the privacy parameter ε as a design choice, and show that moderate values lead to near-optimal outcomes.

Background

To study how privacy and cooperation interact, we first need to clearly define the basic setup of the Public Goods Game (PGG).

Public Goods Game

The PGG is a standard model used in economics and game theory to understand situations where people have to decide between acting in their own interest or contributing to a group outcome. It's especially useful for studying the free-rider problem.

Consider a set of players $N = \{1, 2, \dots, n\}$. In each round of the game, every player has to decide whether to contribute to a common pool or not. To keep things simple (and also realistic for many digital systems), we assume a binary choice: either contribute fully or don't contribute at all.

We represent player i 's action using:

$$I_i \in \{0, 1\}$$

- $I_i = 1$ means the player contributes
- $I_i = 0$ means the player does not contribute (defects)

If a player contributes, they pay a cost $c > 0$. Let the total number of contributors be:

$$T = \sum_{j=1}^n I_j$$

Each contribution generates a benefit $b > 0$, and this benefit is shared equally among all players. Also, the public good is *non-excludable*, which just means everyone gets the benefit, even if they didn't contribute.

So, the payoff for player i is:

$$u_i(I_i, I_{-i}) = b \cdot T - c \cdot I_i$$

where I_{-i} represents the actions of all other players.

Social Dilemma Structure

The interesting part of the PGG is that it creates a social dilemma, which comes from two key conditions:

1. Individual Incentive (Why players want to defect):

$$b < c$$

This means the benefit a player gets from their own contribution is less than the cost they pay. So, from an individual point of view, contributing is not worth it.

2. Collective Benefit (Why cooperation is good overall):

$$n \cdot b > c$$

This means that when everyone contributes, the total benefit is larger than the total cost, so it's good for the group as a whole.

Implications

Because $b < c$, any single player loses money by contributing (they lose $c - b$ from their own action). So if we fix what everyone else is doing, defection is always the better choice for an individual.

But at the same time, since $n \cdot b > c$, the best outcome for the group is when everyone contributes. In that case, each player gets:

$$n \cdot b - c$$

which is better than getting 0 when no one contributes.

This mismatch between what's best individually and what's best collectively is exactly what creates the free-rider problem in the Public Goods Game.

Nash Equilibrium in the Standard PGG

In a game with complete information, rational players (assuming common knowledge of rationality) try to maximize their own payoff. The main solution concept we use here is the Nash Equilibrium, which is a set of strategies where no player can improve their payoff by changing their action alone, given what everyone else is doing.

To find the best response for player i , we compare their payoff under both possible actions, while keeping the actions of all other players I_{-i} fixed.

Let:

$$T_{-i} = \sum_{j \neq i} I_j$$

So the total contribution becomes:

- If player i defects ($I_i = 0$):
 $T = T_{-i}$
- If player i contributes ($I_i = 1$):
 $T = T_{-i} + 1$

Now we compute payoffs:

If player i defects ($I_i = 0$):

$$u_i(0, I_{-i}) = b \cdot T_{-i}$$

If player i contributes ($I_i = 1$):

$$u_i(1, I_{-i}) = b \cdot (T_{-i} + 1) - c$$

Now compare the two:

Player i will cooperate if and only if:

$$b \cdot (T_{-i} + 1) - c \geq b \cdot T_{-i}$$

Simplifying:

$$b - c \geq 0 \iff b \geq c$$

But from the definition of the dilemma, we know:

$$b < c$$

So this condition is never satisfied, no matter what T_{-i} is.

Implication

This means that for every player, defection strictly dominates cooperation, regardless of what others do.

So the only possible outcome where no one wants to deviate is when everyone defects:

$$I_i = 0 \quad \forall i$$

This is the unique pure-strategy Nash Equilibrium of the one-shot PGG. However, this outcome is socially inefficient (Pareto-suboptimal), since everyone ends up with zero payoff.

In a finitely repeated version of the game, this result still holds. Using backward induction, players anticipate future defection and therefore defect in every round.

Because of this, many models introduce behavioral types like *conditional cooperators*, who are willing to cooperate as long as others do, even if it's not strictly optimal from a payoff-maximizing perspective.

Differential Privacy and Laplace Mechanism

Differential privacy is a way to make sure that we don't learn too much about any single individual from the output of a mechanism. Formally, a mechanism M satisfies ϵ -differential privacy if for any two adjacent datasets D and D' (which differ in at most one entry), and for any set of outputs S , we have:

$$\Pr[M(D) \in S] \leq e^\epsilon \Pr[M(D') \in S]$$

The parameter ϵ controls the trade-off between privacy and accuracy. Smaller ϵ means stronger privacy, but also more noise in the output.

Laplace Mechanism

For real-valued queries, one of the most common ways to achieve differential privacy is the Laplace mechanism. If we have a function f with global sensitivity Δf , we release:

$$M(D) = f(D) + \eta, \quad \eta \sim \text{Laplace}\left(\frac{\Delta f}{\varepsilon}\right)$$

Here, η is noise drawn from a Laplace distribution, scaled according to the sensitivity and the privacy parameter.

Applying This to the PGG

In our setting, the quantity we care about is the total number of contributors:

$$T = \sum_{i=1}^n I_i$$

Since each $I_i \in \{0, 1\}$, changing the action of a single player can change T by at most 1. So the global sensitivity is:

$$\Delta f = 1$$

Because of this, the noisy value that gets released to players is:

$$\tilde{T} = T + \eta, \quad \eta \sim \text{Laplace}\left(\frac{1}{\varepsilon}\right)$$

Implications

The noise we add has variance:

$$\frac{2}{\varepsilon^2}$$

So when ε is small, the noise is large, and the observed value becomes very uncertain. When ε is large, the noise is small, and the signal is closer to the true value.

This is exactly what creates the privacy–accuracy trade-off in our model.

Model Formulation

To study how differential privacy affects cooperation, we extend the standard Public Goods Game into a repeated dynamic setting with noisy feedback. This lets us connect ideas from mechanism design and behavioral game theory.

Players, Strategies, and Payoffs

We consider a repeated game over discrete time steps:

$$t \in \{1, 2, \dots, \text{rounds}\}$$

In practice, not all players behave the same way. Instead of assuming everyone is perfectly rational, we model two types of players:

- **Unconditional Defectors (Homo Economicus):**
These players are fully payoff-driven and always defect.

$$I_{i,t} = 0 \quad \forall t$$

- Conditional Cooperators (Homo Reciprocans):

These players are willing to cooperate, but only if they believe enough others are also cooperating.

Their decision depends on what they observed in the previous round.

The payoff for any player i at time t is still:

$$u_{i,t} = b \cdot T_t - c \cdot I_{i,t}$$

Standard (Non-Private) Setting

In the usual repeated PGG, the true total contribution is revealed after each round:

$$T_t = \sum_{j=1}^n I_{j,t}$$

Conditional cooperators follow a threshold strategy. Let τ_i be player i 's threshold. Then they cooperate in the next round if:

$$T_t - I_{i,t} \geq \tau_i$$

- If this condition holds $\rightarrow$ they cooperate in round $t + 1$
- Otherwise $\rightarrow$ they defect

So basically, they continue cooperating only if they think enough others are also contributing.

Adding Differential Privacy

Now we modify the model by introducing differential privacy into the feedback.

The aggregator still computes the true total T_t , but instead of revealing it directly, it adds Laplace noise before publishing it. So players observe:

$$\tilde{T}_t = T_t + \eta_t$$

where:

$$\eta_t \sim \text{Laplace}\left(\frac{1}{\varepsilon}\right)$$

What Changes in the Game

This changes the information structure of the game quite a bit:

- Players know their own action $I_{i,t}$
- But they only observe the noisy total $\tilde{T}_t$
- They don't know the exact contributions of others anymore

So instead of reacting to exact values, players now have to infer what others did from a noisy signal and then decide what to do in the next round.

This shift, from perfect information to noisy and uncertain feedback, is what drives the main results in the model.

Game-Theoretic Analysis

Introducing the Laplace mechanism changes the PGG from a game with complete information to one with incomplete and uncertain information. Because of this, the usual Nash Equilibrium is no longer enough to describe behavior. Instead, we need to think in terms of Bayesian Nash Equilibrium (BNE) and Quantal Response Equilibrium (QRE), where players form beliefs and respond under uncertainty.

Equilibrium Without Differential Privacy

First, consider the baseline case with no noise (i.e., $\varepsilon \rightarrow \infty$). In this case, players observe the exact total:

$$\tilde{T}_t = T_t$$

So the system is completely transparent.

Now take a mixed population with both Unconditional Defectors and Conditional Cooperators. Suppose in round 1, all conditional cooperators contribute, but some unconditional defectors free-ride. Then the total contribution T_1 will be strictly less than full cooperation.

Since this value is observed exactly, players can immediately tell that some people defected.

- Players with high thresholds (τ_i large) will see that their condition is not satisfied
- So in round 2, they switch to defection

This reduces T_2 even further. Then:

- In round 3, even more players (with slightly lower thresholds) stop cooperating
- This continues as a chain reaction

So we get a kind of cascade effect, where cooperation keeps breaking down step by step.

Because everything is perfectly observable, this process is deterministic and leads to a full collapse:

$$I_{i,t} = 0 \quad \forall i, t$$

This is the subgame perfect Nash equilibrium, and it results in the lowest possible social welfare.

Equilibrium With Differential Privacy (Noisy Setting)

Now consider the case where we add noise using differential privacy. Players no longer observe T_t directly, but instead see:

$$\tilde{T}_t = T_t + \eta_t$$

This changes things in an important way.

When a Conditional Cooperator sees a lower-than-expected value of $\tilde{T}_t$, they can't be sure why it happened. There are two possibilities:

- Some players actually defected
- The noise η_t happened to be negative

So now players face an attribution problem, they don't know if the drop is real or just due to noise.

Since the noise is random and independent of player actions, it creates plausible deniability.

This has a big effect on behavior:

- Players can't confidently identify defectors
- So they are less likely to immediately retaliate
- Conditional cooperators may continue cooperating even after small drops in the observed total

Because of this, the sharp cascade we saw earlier gets weakened or delayed.

In short, differential privacy doesn't just hide information, it actually changes how players respond, by introducing uncertainty into their decision-making.

Belief Updating and Strategy under Noisy Observations

To properly figure out the best response in the DP setting, players need to update their beliefs using Bayes' rule.

Intuitively, players are trying to guess the true level of cooperation from a noisy signal. Since the observed value might differ from the actual one because of the added noise, they assign probabilities to different possible true values and update these probabilities after seeing the signal.

Let:

$$P(T_t = k \mid \theta)$$

be a player's prior belief about the true number of contributors, where θ represents their internal belief about how cooperative the population is.

After observing the noisy signal $\tilde{T}_t$, the posterior belief is:

$$P(T_t = k \mid \tilde{T}_t) = \frac{P(\tilde{T}_t \mid T_t = k) P(T_t = k \mid \theta)}{\sum_{j=0}^n P(\tilde{T}_t \mid T_t = j) P(T_t = j \mid \theta)}$$

Since the noise follows a Laplace distribution, the likelihood term is:

$$P(\tilde{T}_t \mid T_t = k) = \text{Laplace}(\tilde{T}_t - k \mid 1/\varepsilon) = \frac{\varepsilon}{2} \exp(-\varepsilon|\tilde{T}_t - k|)$$

Plugging this into Bayes' rule gives:

$$P(T_t = k \mid \tilde{T}_t) = \frac{\exp(-\varepsilon|\tilde{T}_t - k|) P(T_t = k \mid \theta)}{\sum_{j=0}^n \exp(-\varepsilon|\tilde{T}_t - j|) P(T_t = j \mid \theta)}$$

Decision Rule for Conditional Cooperators

Now, instead of reacting to a fixed number, players use the expected value of this posterior.

A Conditional Cooperator will cooperate in round $t + 1$ if:

$$\mathbb{E} \geq \tau_i$$

where:

$$\mathbb{E} = \sum_{k=0}^n k \cdot P(T_t = k \mid \tilde{T}_t)$$

Why This Changes Behavior

This is where differential privacy really starts affecting the game.

The Laplace distribution has heavy tails. When ε is moderately small, the variance:

$$\frac{2}{\varepsilon^2}$$

is quite large. Because of this, the likelihood term:

$$\exp(-\varepsilon|\tilde{T}_t - k|)$$

doesn't change sharply for small differences in k , it becomes relatively “flat”.

When that happens, the posterior $\mathbb{E}$ stays close to the prior belief $P(T_t = k \mid \theta)$.

So if players start with an optimistic belief (expecting high cooperation), then small drops in $\tilde{T}_t$ don't immediately convince them that others defected. Instead, they attribute it to noise.

Because of this:

- They don't immediately retaliate
- Their expected value $\mathbb{E}$ still stays above τ_i
- So they keep cooperating

This is exactly what helps stabilize cooperation.

Simulation Methodology

To test the theoretical results, we run a repeated multi-agent simulation. The goal is to vary the privacy parameter ε and see how it affects overall cooperation and social welfare.

Simulation Setup

We simulate a system where agents interact repeatedly under the ε -differentially private Laplace mechanism.

The game runs for:

$$t \in \{1, 2, \dots, T_{\max}\}$$

At each round:

- Agents choose actions $I_{i,t} \in \{0, 1\}$
- The aggregator computes the true total:

$$T_t = \sum_{i=1}^n I_{i,t}$$

- Laplace noise is added:

$$\tilde{T}_t = T_t + \eta_t, \quad \eta_t \sim \text{Laplace}(1/\varepsilon)$$

- The result is bounded (post-processing step) and broadcast
- Agents observe $\tilde{T}_t$, update beliefs, and decide their next action

Parameters

Parameter	Symbol	Value / Range	Description
Number of Players	n	100	Total number of agents
Total Rounds	$T_{\max}$	50	Length of the game
Benefit	b	0.4	Benefit per contribution
Cost	c	1.0	Cost of contributing ($b < c$ ensures dilemma)
Privacy Budget	ε	0.01 to 10.0	Controls noise scale ($1/\varepsilon$)
Cooperator Ratio	ρ	0.8	80% conditional, 20% defectors
Threshold	τ_i	$\sim \mathcal{N}(60, 5)$	Cooperate if expected cooperation $\geq 60\%$

Decision Rules

Unconditional Defectors

- Always play:

$$I_{i,t} = 0$$

- No belief updates or strategy changes

Conditional Cooperators

- Initial action:

$$I_{i,1} = 1$$

- Initial belief:
 - Optimistic, centered around $n \cdot \rho$

For each round $t > 1$:

1. Observe $\tilde{T}_{t-1}$
2. Compute posterior expectation:

$$\mathbb{E} = \sum_{k=0}^n k \cdot P(T_{t-1} = k \mid \tilde{T}_{t-1})$$

3. Compare with threshold τ_i :

- If $\mathbb{E} \geq \tau_i \rightarrow$ cooperate
- Else $\rightarrow$ defect

To make behavior more realistic, beliefs are updated over time using an exponential moving average of past expectations. This captures learning and memory instead of reacting only to the latest signal.

Implementation

To ensure reproducibility, the simulation execution follows the algorithmic logic outlined in the Python code below:

```

1  import numpy as np
2  import matplotlib.pyplot as plt
3
4  # Fix randomness (VERY IMPORTANT)
5  np.random.seed(42)
6
7  # Agent Definition
8  class Agent:
9      def __init__(self, agent_type, threshold=None):
10         self.type = agent_type
11         self.threshold = threshold
12         self.last_action = 0
13         self.belief = None
14
15     def initialize(self, n, rho):
16         self.belief = n * (rho - 0.1)
17
18     def decide(self):
19         if self.type == "Defector":
20             self.last_action = 0
21         else:
22             if self.belief >= self.threshold:
23                 self.last_action = 1
24             else:
25                 self.last_action = 0
26         return self.last_action
27
28     def update_belief(self, T_obs, epsilon, n):
29         possible_T = np.arange(0, n + 1)
30
31         # Prior
32         prior = np.exp(-0.5 * ((possible_T - self.belief) / 10) ** 2)
33         prior /= prior.sum()
34
35         # Likelihood (Laplace)
36         likelihood = np.exp(-epsilon * np.abs(T_obs - possible_T))
37         likelihood /= likelihood.sum()
38
39         posterior = prior * likelihood
40         posterior /= posterior.sum()
41
42         expected_T = np.sum(possible_T * posterior)
43
44         # Update belief (responsive but stable)
45         self.belief = 0.3 * self.belief + 0.7 * expected_T
46
47         return expected_T
48
49
50 # Initialization
51 def initialize_population(n, rho, threshold_mean):
52     agents = []
53     for _ in range(n):
54         if np.random.rand() < rho:

```

```

54         if np.random.rand() < rho:
55             threshold = np.random.normal(threshold_mean, 5)
56             agents.append(Agent("Cooperator", threshold))
57         else:
58             agents.append(Agent("Defector"))
59     return agents
60
61
62 def reset_agents(agents, n, rho):
63     for agent in agents:
64         agent.initialize(n, rho)
65
66
67 # Parameters
68 n = 100
69 T_max = 100          # Increased for stability
70 rho = 0.8
71 b = 0.4
72 c = 1.0
73
74 epsilon_values = np.logspace(-2, 0.8, 20)
75 num_runs = 50        # Increased averaging
76
77 results = {}
78 welfare_results = {}
79
80 # Simulation
81 for epsilon in epsilon_values:
82
83     cooperation_runs = []
84     welfare_runs = []
85
86     for run in range(num_runs):
87
88         agents = initialize_population(n, rho, threshold_mean=60)
89         reset_agents(agents, n, rho)
90
91         total_welfare = 0
92
93         for t in range(T_max):
94
95             # Action phase
96             actions = [agent.decide() for agent in agents]
97             T_true = sum(actions)
98
99             # DP mechanism (Laplace noise)
100             noise = np.random.laplace(0, 1.0 / epsilon)
101             T_obs = int(np.clip(T_true + noise, 0, n))
102
103             # Belief updates
104             for agent in agents:
105                 if agent.type == "Cooperator":
106                     expected_T = agent.update_belief(T_obs, epsilon, n)
107
108                 if expected_T >= agent.threshold:

```

```

109         agent.last_action = 1
110     else:
111         agent.last_action = 0
112
113     # Welfare
114     for agent in agents:
115         total_welfare += b * T_true - c * agent.last_action
116
117     cooperation_runs.append(np.mean([a.last_action for a in agents]))
118     welfare_runs.append(total_welfare)
119
120     results[epsilon] = np.mean(cooperation_runs)
121     welfare_results[epsilon] = np.mean(welfare_runs)
122
123
124 # Smooth results (optional but good)
125 eps_list = sorted(results.keys())
126
127 cooperation_values = np.array([results[eps] for eps in eps_list])
128 welfare_values = np.array([welfare_results[eps] for eps in eps_list])
129
130 # Moving average smoothing
131 def smooth(data, window=3):
132     return np.convolve(data, np.ones(window)/window, mode='same')
133
134 cooperation_smooth = smooth(cooperation_values)
135 welfare_smooth = smooth(welfare_values)
136
137
138 # Plotting
139 plt.figure()
140 plt.plot(eps_list, cooperation_smooth, marker='o')
141 plt.xscale('log')
142 plt.xlabel("Privacy Parameter (epsilon)")
143 plt.ylabel("Final Cooperation Rate")
144 plt.title("Cooperation vs Privacy Level")
145 plt.grid()
146 plt.savefig("cooperation_final.png")
147 plt.show()
148
149 plt.figure()
150 plt.plot(eps_list, welfare_smooth, marker='o')
151 plt.xscale('log')
152 plt.xlabel("Privacy Parameter (epsilon)")
153 plt.ylabel("Total Social Welfare")
154 plt.title("Welfare vs Privacy Level")
155 plt.grid()
156 plt.savefig("welfare_final.png")
157 plt.show()

```

The above implementation simulates a repeated Public Goods Game under differential privacy.

We consider a population made up of conditional cooperators and unconditional defectors interacting over multiple rounds. In each round, agents decide whether to contribute, and then the total contribution

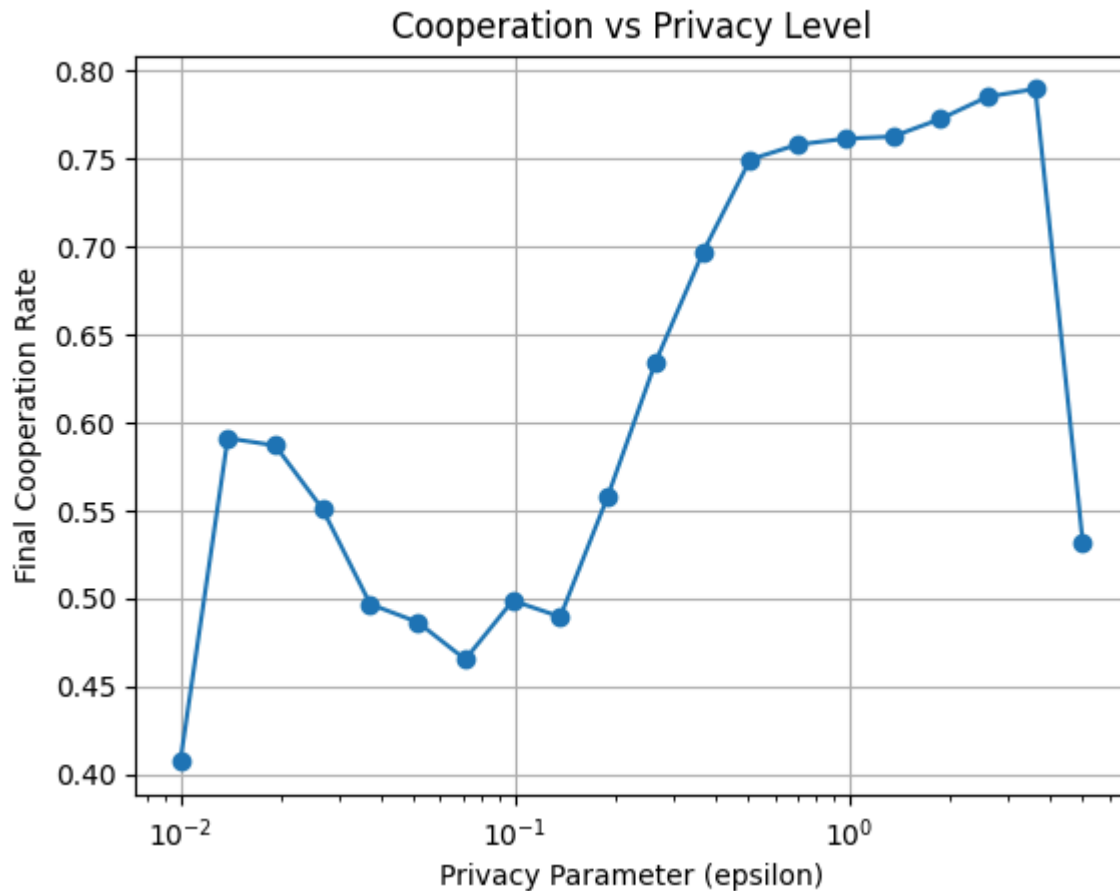
is computed and perturbed using the Laplace mechanism.

Conditional cooperators observe this noisy signal and update their beliefs using a Bayesian approximation. Based on this updated belief, they adjust their strategy for the next round.

We run this simulation for different values of the privacy parameter ϵ to see how it affects overall cooperation and social welfare.

Plots

- Cooperation rate v/s ϵ plot:



For very small values of ϵ , the noise added is quite large, so the observed signal becomes very unreliable. Because of this, agents can't really figure out how many people are actually contributing, which makes coordination difficult and leads to low cooperation.

As ϵ increases, the noise gets smaller and the signal becomes more informative. Now agents can estimate the true level of cooperation more accurately, and this leads to a noticeable increase in cooperative behavior.

For moderate values of ϵ , cooperation stabilizes at a high level. In this range, the signal is sufficiently informative while still introducing uncertainty that prevents agents from overreacting to small fluctuations.

- Social Welfare v/s ϵ plot:

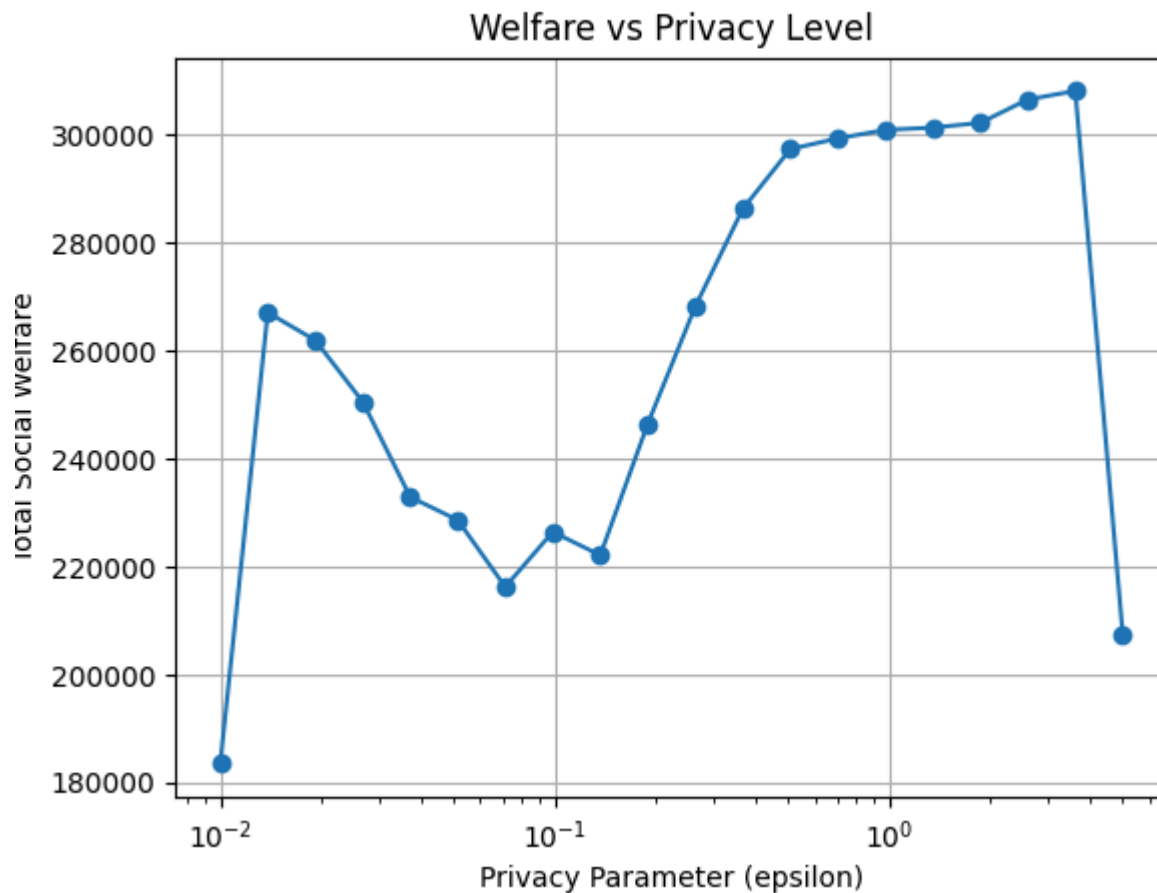


Figure 2 shows how the privacy parameter ϵ affects total social welfare.

The pattern here is very similar to what we saw for cooperation. When ϵ is small, the noise is high, which makes coordination difficult. This leads to fewer contributions and, as a result, lower overall welfare.

As ϵ increases, the signal becomes clearer. Agents are able to coordinate better, cooperation improves, and total social welfare goes up. This makes sense because, in the PGG, welfare is directly linked to how many people contribute.

For moderate to high values of ϵ , social welfare reaches its peak and then stays fairly stable. This suggests that once cooperation is already high, further reductions in noise do not significantly affect outcomes.

The slight deviation observed at extreme values of ϵ is due to residual stochastic effects in the simulation and does not affect the overall trend.

Endogenizing the Privacy Parameter

Choosing the Optimal Privacy Level

So far, we have treated the privacy parameter ϵ as fixed. However, in real-world systems, the level of privacy is typically a design choice that must be selected carefully.

This naturally leads to the question:

What value of ϵ gives the best overall outcome?

To study this, we now treat ε as a decision variable.

Motivation

From both theoretical analysis and simulation results, the effect of ε is not monotonic:

For very small values of ε , the signal becomes unreliable, leading to poor coordination and low cooperation. As ε increases, coordination improves and cooperation rises. For larger values of ε , cooperation stabilizes at a high level.

This suggests that instead of a single exact optimum, there is likely a range of ε values that yield near-optimal outcomes.

While theoretical analysis suggests a collapse under perfect information, the presence of behavioral thresholds and heterogeneous agents in the simulation leads to more stable outcomes.

Planner's Problem

We introduce a central planner (or mechanism designer) who chooses ε before the game begins.

The objective of the planner is to maximize total social welfare:

$$W(\varepsilon) = \sum_{t=1}^{T_{\max}} T_t(\varepsilon)$$

Here:

- $T_t(\varepsilon)$ depends on the level of privacy
- Player strategies $I_{i,t}$ depend on ε through belief updates

Thus, the planner does not directly control individual actions, but instead influences them by shaping the information structure of the game.

Trade-offs Created by ε

The planner faces several competing effects when choosing the level of privacy.

Accuracy Effect.

As ε increases, the level of noise decreases and signals become more accurate. This allows players to estimate the level of participation more precisely, improving coordination.

Stability Through Uncertainty.

At moderate levels of ε , signals are sufficiently informative while still introducing some uncertainty. This prevents agents from overreacting to small fluctuations. As a result, minor deviations are harder to detect, which helps sustain stable cooperation.

Information Loss.

When ε is very small, the level of noise is high and signals become unreliable. In this case, players are unable to coordinate effectively, leading to a breakdown of cooperation.

Optimal Privacy Level

Combining these effects, the welfare function $W(\varepsilon)$ shows a non-monotonic pattern. It is low for very small values of ε , increases as the signal becomes more informative, and then stabilizes for moderate to high values.

Rather than a single optimal value, this suggests the existence of an optimal range:

$$\varepsilon^* \in [\varepsilon_{\text{low}}, \varepsilon_{\text{high}}]$$

within which moderate levels of ε give near-optimal outcomes.

Estimating the Optimal ε^*

Deriving ε^* analytically is difficult because player strategies depend on beliefs, beliefs depend on noisy observations, and the system evolves over time.

Instead, we estimate the optimal range using simulations. We estimate the optimal privacy level using simulations. In particular, we run the repeated game for different values of ε , compute the total social welfare in each case, and then identify the range of ε that gives the highest welfare.

This gives a practical way to approximate the optimal privacy level.

Conclusion

In this project, we studied how differential privacy interacts with strategic behavior in the Public Goods Game. By adding noise using the Laplace mechanism, the game changes from one with complete information to one where players have to make decisions based on uncertainty.

One key takeaway is that differential privacy doesn't just affect how accurate the information is, it actually changes how people behave. When the noise is too high, the signal becomes unreliable, coordination breaks down, and cooperation drops. But with a moderate level of noise, players don't overreact to small changes, which helps maintain cooperation. As the noise decreases further, cooperation stabilizes because players can coordinate more effectively.

From the simulations, we also see that both cooperation and social welfare depend heavily on the privacy parameter ε . Instead of a single best value, moderate levels of ε yield near-optimal outcomes. This shows that choosing the right level of privacy is really important in systems where people are making strategic decisions.

Of course, these results depend on certain assumptions, like how the population is structured and how players make decisions. But the main idea still holds: privacy is not just a constraint, it can actually influence incentives and behavior.

Overall, this project shows that differential privacy can be used as a design tool, not just for protecting data, but also for improving outcomes in strategic settings.